## Cascaded nonlinearity in metasurface

In this note , I explore the possibility of using cascaded second-order nonlinearity to achieve an effective third-order nonlinearity in a resonant structure. The key lies in designing a metasurface that exhibits resonances at two frequencies, $\omega_1$ and $\omega_2$, where $\omega_2$ is slightly detuned from $2\omega_1$. It is crucial for the second mode to possess the highest quality factor (Q) because it does not need to be coupled to the outside environment.

First let us define the two modes – for fundamental and SH as $f_1(\boldsymbol{r})$ and $f_2(\boldsymbol{r})$ with resonant frequencies $\omega_1$ and $\omega_2 = 2\omega_1 + \Delta\omega$ while their damping rates are $\gamma_{1,2}$. That means that two modes satisfy wave equations

$$\nabla^2 f_{1,2} = -\frac{n^2\omega_{1,2}^2}{c^2} \tag{1}$$

The electric fields are $E_1 = A_1(t) f_1(\boldsymbol{r}) e^{-j\omega t}$ and $E_2 = A_2(t) f_2(\boldsymbol{r}) e^{-j2\omega t}$ The modes are normalized $\int f_{1,2}^2 d\boldsymbol{r} = 1$. The energies in the modes are

$$u_{1,2} = \frac{\varepsilon_0 n^2}{2} | A_{1,2} |^2 = | a_{1,2} |^2 \tag{2}$$

Now we have nonlinear polarization at frequencies $2\omega$ and $\omega$

$$\begin{aligned} P_{non2} &= \varepsilon_0 \chi^{(2)} E_1^2 e^{-2j\omega t} = \frac{2}{n^2}\chi^{(2)} a_1^2 f_1^2 e^{-2j\omega t} \\ P_{non1} &= \varepsilon_0 \chi^{(2)} E_2 E_1^* e^{-j\omega t} + \varepsilon_0 \chi^{(3)} E_1^2 E_1^* e^{-j\omega t} = \frac{2}{n^2}\chi^{(2)} a_2 a_1^* f_1 f_2 e^{-j\omega t} + \frac{2^{3/2}}{n^3}\varepsilon_0^{3/2}\chi^{(3)} a_1^2 a_1^* e^{-j\omega t} f_1^3 \end{aligned} \tag{3}$$

Substitute it into wave equations

$$\begin{aligned} \nabla^2 E_2 - \frac{n^2}{c^2}\frac{\partial^2 E_2}{\partial t^2} &= \frac{1}{\varepsilon_0}\frac{\partial^2 P_{non,2}}{\partial t^2} \\ \nabla^2 E_1 - \frac{n^2}{c^2}\frac{\partial^2 E_2}{\partial^2 t^2} &= \frac{1}{\varepsilon_0}\frac{\partial^2 P_{non,1}}{\partial t^2} \end{aligned} \tag{4}$$

After getting read of second derivatives of $A_{1,2}$ we are left with

$$\begin{aligned} -\frac{n^2}{c^2}(\omega_2^2 - 4\omega^2) A_2 f_2 + 4j\omega \frac{n^2}{c^2}\frac{\partial A_2}{\partial t} f_2 &= -\frac{4\omega^2}{c^2}\chi^{(2)} A_1^2 f_1^2 \\ -\frac{n^2}{c^2}(\omega_1^2 - \omega^2) A_1 f_1 + 2j\omega \frac{n^2}{c^2}\frac{\partial A_2}{\partial t} f_1 &= -\frac{\omega^2}{c^2}\chi^{(2)} A_2 A_1^* f_1 f_2 - \frac{\omega^2}{c^2}\chi^{(3)} A_1^2 A_1^* f_1^3 \end{aligned} \tag{5}$$

Do some cancellations and also usual think with multiplying by $f_{1,2}$ and subsequent integration

$$
\begin{aligned}
&-(\omega_2 - 2\omega)A_2 + j\frac{\partial A_2}{\partial t} = -\frac{\omega}{n^2}A_1^2\int \chi^{(2)} f_1^2 f_2 d\boldsymbol{r} \\
&-(\omega_1 - \omega)A_2 + j\frac{\partial A_2}{\partial t} = -\frac{\omega}{2n^2}A_2 A_1^*\int \chi^{(2)} f_1^2 f_2 d\boldsymbol{r} - \frac{\omega}{2n^2}A_1^2 A_1^*\int \chi^{(3)} f_1^4 d\boldsymbol{r}
\end{aligned} \tag{6}
$$

And so

$$
\begin{aligned}
&\frac{da_2}{dt} = -j\Delta\omega a_2 + j\frac{\omega}{n^3}\sqrt{\frac{2}{\varepsilon_0}}a_1^2\int \chi^{(2)} f_1^2 f_2 d\boldsymbol{r} \\
&\frac{da_2}{dt} = -j(\omega_1 - \omega)a_1 + j\frac{\omega}{2n^3}\sqrt{\frac{2}{\varepsilon_0}}a_2 a_1^*\int \chi^{(2)} f_1^2 f_2 d\boldsymbol{r} + j\frac{\omega}{2n^3}\frac{2}{\varepsilon_0}a_1^2 a_1^*\int \chi^{(3)} f_1^4 d\boldsymbol{r}
\end{aligned} \tag{7}
$$

Let us introduce effective coupling coefficient

$$
\kappa = \frac{\omega}{n^3}\sqrt{\frac{2}{\varepsilon_0}}\chi_{\max}^{(2)}\int F_2 f_1^2 f_2 d\boldsymbol{r} \quad J^{-1/2}s^{-1} \tag{8}
$$

where $\chi^{(2)}(\boldsymbol{r}) = \chi_{\max}^{(2)} F_2(\boldsymbol{r})$,i.e. $F_2(\boldsymbol{r})$ is normalized spatial distribution of $\chi^{(2)}$, hopefully involving change of sign achieved by poling or some other symmetry breaking arrangement required to make the overlap integral non zero. and also third order coupling

$$
\kappa_3 = \frac{\omega}{n^3\varepsilon_0}\chi_{\max}^{(3)}\int F_3 f_1^4 d\boldsymbol{r} \quad J^{-1}s^{-1} \tag{9}
$$

And now we add in-coupling and radiative and nonradiative loss

$$
\begin{aligned}
&\frac{da_2}{dt} = -j\Delta\omega a_2 - \frac{\gamma_2}{2}a_2 + j\kappa a_1^2 \\
&\frac{da_1}{dt} = -j(\omega_1 - \omega)a_1 - \frac{\gamma_1}{2}a_1 + j\frac{\kappa}{2}a_2 a_1^* + j\kappa_3|a_1|^2 a_1 + j\sqrt{\gamma_{rad}}a_{in}
\end{aligned} \tag{10}
$$

With that…..we just find steady state solution of the first equation

$$
a_2 = \frac{j\kappa a_1^2}{j\Delta\omega + \gamma_2/2} = \frac{\kappa a_1^2}{\Delta\omega - j\gamma_2/2} \approx \frac{\kappa a_1^2}{\Delta\omega} + \frac{j\gamma_2/2}{\Delta\omega^2}\kappa a_1^2 \tag{11}
$$

And substitute it into the second one

$$
\frac{da_1}{dt} = -j(\omega_1 - \omega)a_1 - \frac{\gamma_1}{2}a_1 - \frac{\kappa^2\gamma_2}{4\Delta\omega^2}|a_1|^2 a_1 + j\left(\kappa_3 + \frac{\kappa^2}{2\Delta\omega}\right)|a_1|^2 a_1 + j\sqrt{\gamma_{rad}}a_{in} \tag{12}
$$

So now we have additional term for cascaded nonlinearity

$$\kappa_{3,casc} = \frac{\kappa^2}{2\Delta\omega} = \frac{\omega}{n^6}\frac{\omega}{\Delta\omega}\frac{1}{\varepsilon_0}\left(\chi_{\max}^{(2)}\right)^2 \left|\int F_2 f_1^2 f_2 d\boldsymbol{r}\right|^2$$

And using (9) we obtain effective cascaded nonlinear susceptibility

$$\chi_{casc}^{(3)} = \frac{\kappa_{3,casc} n^3 \varepsilon_0}{\omega \int F_3 f_1^4 d\boldsymbol{r}} = \frac{\omega}{\Delta\omega}\frac{1}{n^3}\left(\chi_{\max}^{(2)}\right)^2 \frac{\left|\int F_2 f_1^2 f_2 d\boldsymbol{r}\right|^2}{\int F_3 f_1^4 d\boldsymbol{r}} \tag{13}$$

Let us now ascertain what do we get. The key here is mode matching overlap term which require some kind of quasi-phase matching by variation of $F_2$ to make sure the overlap integral is not zero. Let us assume that we can get it as large as 10%. Second, we assumed that $\Delta\omega >> \gamma_2 / 2$ so we can say that $\omega / \Delta\omega \sim Q_2 / 4$ or something to this extent. Let us assume this factor is 1000. Now, assume lithium niobate then $n^3 \sim 10$. Then

$$\chi_{casc}^{(3)} \sim 10\left(\chi_{\max}^{(2)}\right)^2 \sim 10^{-20} m^2 / V^2 \tag{14}$$

This is higher than the native third order nonlinearity of lithium niobate $\chi_{casc}^{(3)} \sim 10^{-21} m^2 / V^2$

→ Phase modulator through $\chi^{(2)}$ ☆
→ Faster, larger than $\chi^{(3)}$
→ Overtaxing